Supplemental Material: The Bullet Cluster Mass–Gas Offset as an Independent Test of $c_s^2 = 0$ in Khronon Condensation

Sheng-Kai Huang[1, *]

[1] *Independent Researcher*

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The Bullet Cluster (1E 0657-558) provides a direct test of dark matter collisionlessness: lensing maps trace the dark matter to a position offset by $\sim$720 kpc/h from the X-ray-emitting gas, while the gas is decelerated by ram pressure. For Khronon condensation with $c_s^2 = 0$ (BS2025 Khronon–Tensor regime), the sound-crossing time over the cluster scale is infinite, the condensate behaves as pressureless collisionless dust, and the lensing–gas offset is reproduced trivially. For pure scalar Khronon constrained by CMB ($c_s^2 \lesssim 3.2 \times 10^{-6}$, TKS2016 [?]), the sound-crossing time over 1 Mpc is $\sim$ 1800 Myr, about 9× longer than the $\sim$ 200 Myr collision crossing. The offset is preserved but marginally so; the $c_s^2 = 0$ limit is preferred. The Bullet Cluster therefore provides a sixth *semi-quantitative* test of the $c_s^2 = 0$ prediction, weaker than the CMB bound (which already requires $c_s^2 \lesssim 3.2 \times 10^{-6}$) but consistent and independent.

I. THE BULLET CLUSTER PHENOMENON

[KNOWN] The Bullet Cluster, 1E 0657-558 at $z = 0.296$, is a major-merger system in which two galaxy clusters have recently collided. Chandra X-ray imaging [? ?] and weak gravitational lensing maps [? ?] together establish three observational facts:

1. the dominant baryonic mass component (intracluster gas, $\sim$ 90% of the baryons) is decelerated by ram pressure during the collision and lags behind the galaxies by approximately 720 kpc/h_{70};

2. the lensing convergence map peaks coincide with the galaxy distributions, not the gas, with a significance of 8σ;

3. the inferred mass-to-light ratio at the lensing peaks rules out modified-gravity-only explanations without an additional collisionless mass component [? ?].

The collision velocity inferred from hydrodynamical simulations is $v_{\rm coll} \approx 4700$ km/s [?]. The collision crossing time over the cluster scale ($L \sim 1$ Mpc) is therefore

$$t_{\rm coll} = \frac{L}{v_{\rm coll}} \approx \frac{3.09 \times 10^{22}\ {\rm m}}{4.7 \times 10^6\ {\rm m/s}} \approx 6.6 \times 10^{15}\ {\rm s} \approx 210\ {\rm Myr}. \tag{1}$$

II. KHRONON CONDENSATE AS COLLISIONLESS DUST

[DERIVED] At the ghost condensation point, the BS kinetic function $K(Q) = \mu^2(Q-1)^2$ has $K'(Q_0) \to 0$ on the cosmological background (Paper 3, Sec. III). The k-essence sound speed

$$c_s^2 = \frac{K'(Q_0)}{K'(Q_0) + 2Q_0 K''(Q_0)} \to 0 \tag{2}$$

vanishes exactly in the Khronon–Tensor formulation [?] where the auxiliary tensor field absorbs the residual pressure perturbation. In the pure scalar Khronon case (without the tensor field), the CMB constrains $c_s^2 < 3.21 \times 10^{-6}$ at 99.7% CL (TKS2016 [?]).

For a pressureless or near-pressureless fluid, the Euler equation

$$\dot{\mathbf{v}}_K + (\mathbf{v}_K \cdot \nabla)\mathbf{v}_K = -\nabla\Phi - \frac{c_s^2}{\rho_K}\nabla\rho_K \tag{3}$$

loses its pressure-gradient term in the $c_s^2 \to 0$ limit. The condensate then evolves freely under gravity alone, exactly like collisionless cold dark matter. In a cluster collision the Khronon component (i) does not develop a shock front, (ii) does not transfer momentum to or from the baryonic gas, and (iii) passes through the collision region in straight (geodesic) trajectories, only weakly deflected by the gravitational potential of the smaller cluster.

III. QUANTITATIVE OFFSET PRESERVATION

A. Sound-crossing time

[DERIVED] The sound-crossing time of the Khronon condensate over a length scale L is

$$t_{\rm sc}(L) = \frac{L}{c_s} = \frac{L}{c\sqrt{c_s^2/c^2}}. \tag{4}$$

For pressure perturbations to redistribute mass over a cluster scale during the collision, we require $t_{\rm sc}(L) \lesssim t_{\rm coll}$, i.e.

$$\frac{c_s^2}{c^2} \gtrsim \left(\frac{v_{\rm coll}}{c}\right)^2 = \left(\frac{4700}{3 \times 10^5}\right)^2 \approx 2.5 \times 10^{-4}. \tag{5}$$

* akai@fawstudio.com

Conversely, for the offset to be preserved at the ~ 720 kpc level over the ~ 200 Myr post-collision interval, we need

$$\frac{c_s^2}{c^2} \ll 2.5 \times 10^{-4}. \tag{6}$$

B. Comparison with CMB and $c_s^2 = 0$

[NEW SYNTHESIS] The CMB-allowed range $c_s^2/c^2 \lesssim 3.21\times10^{-6}$ from TKS2016 is roughly two orders of magnitude tighter than the Bullet Cluster requirement (**??**). In that case, the Khronon sound-crossing time over 1 Mpc is

$$t_{sc}^{(CMB)} = \frac{L}{c\sqrt{3.21\times10^{-6}}} = \frac{3.09\times10^{22}}{3\times10^{8}\cdot 1.79\times10^{-3}}\ \mathrm{s} \approx 1.8\times10^{3}\ \mathrm{Myr}, \tag{7}$$

about $9\times$ longer than t_{coll}. At this rate the Khronon would diffuse only $\Delta x \sim c_s t_{coll} \approx 540$ km/s $\times$ 200 Myr $\approx$ 110 kpc, i.e. a small fraction of the observed 720 kpc offset. The offset is preserved but only marginally; in particular, the offset's sharp edge in the convergence map (deviation < 100 kpc) is more naturally explained by $c_s^2 = 0$ exactly.

For the BS2025 Khronon–Tensor case, $c_s^2 = 0$ identically, $t_{sc} = \infty$, and no diffusion of the lensing peak occurs. This is the limit favored by the data.

IV. STATUS AS AN INDEPENDENT TEST

[NEW SYNTHESIS] The Bullet Cluster offset places a constraint

$$c_s^2/c^2 \lesssim 10^{-4} \quad \text{(order of magnitude)}, \tag{8}$$

which is *independent* of the CMB bound (the two probes use entirely disjoint physics: weak lensing plus X-ray hydrodynamics versus the acoustic peaks of the temperature power spectrum). Equation (**??**) is, however, weaker than the TKS2016 CMB bound of 3.21×10^{-6} by a factor ~ 30.

We therefore take the Bullet Cluster as *semi-quantitative supporting evidence* for the $c_s^2 = 0$ regime: it confirms that the Khronon condensate behaves as collisionless matter on megaparsec scales, but the constraint it places on c_s^2 is not new information beyond the CMB. The strength of this argument is in its independence: the Khronon framework predicts the same sound speed ($c_s^2 = 0$) governs both the CMB acoustic oscillations *and* the cluster-merger phenomenology, and both observations are consistent with this single prediction. A failure of either test would falsify the framework.

The same argument applies to other dissociated clusters: A520 (the "Train Wreck") [**?**], MACS J0025.4-1222 [**?**], and DLSCL J0916.2+2951 (the "Musketball") [**?**]. Of these, A520 has been argued to require collisional dark matter; we note that within the Khronon framework, $c_s^2 = 0$ is a hard prediction and any confirmed $c_s^2 > 10^{-4}$ from cluster mergers would falsify the BS2025 Khronon–Tensor formulation. A520's status remains contested [**?**] and we do not interpret it as falsifying.

TABLE I. Sound-crossing time over $L = 1$ Mpc for varying c_s^2, compared to $t_{coll} \approx 210$ Myr. "Diffusion" is the implied positional drift of Khronon mass during the collision.

c_s^2/c^2	t_{sc}	Diffusion
0 (BS2025)	∞	0
10^{-8}	33000 Myr	$\lesssim 6$ kpc
3.21×10^{-6} (CMB)	1800 Myr	~ 110 kpc
10^{-4}	330 Myr	~ 600 kpc (offset diffuses)
2.5×10^{-4} (Bullet bound)	210 Myr	~ 1 Mpc (excluded)

V. COMPARISON WITH SIMULATIONS

The Bullet Cluster has been extensively simulated in the standard CDM framework [**?** **?** **?**]. Cosmological hydrodynamics simulations with collisionless particle-DM and SPH gas reproduce the ~ 720 kpc offset, the X-ray morphology, and the lensing convergence peaks given $v_{coll} \approx 4500$–4900 km/s. At the level of cluster dynamics, a $c_s^2 = 0$ Khronon condensate is dynamically *indistinguishable* from collisionless particle CDM: the equation of state is $w \approx 0$ on sub-cosmological scales, the pressure gradient vanishes, and the only coupling to baryons is via gravity.

The distinction would appear at the *interior microscopics*: collisionless particle CDM has a velocity-dispersion-supported phase-space density, while the Khronon condensate is a classical scalar field with a fixed value of $\delta = Q-1$. In principle, this might affect core profiles in the densest cluster regions (the inner ~ 100 kpc); a more careful study of strong-lensing cores in dissociated clusters would be needed to test this, and we leave it to future work.

VI. SUMMARY

For Khronon condensation with $c_s^2 = 0$ (Khronon–Tensor, BS2025) the Bullet Cluster mass–gas offset is reproduced naturally, with the condensate behaving as pressureless collisionless dust on cluster-merger timescales. The sound-crossing time over 1 Mpc is $t_{sc} = \infty$, and the lensing peak suffers no diffusion during the ~ 200 Myr collision. This adds the Bullet Cluster as an independent (sixth) consistency check of the $c_s^2 = 0$ prediction listed in Paper 3. The constraint $c_s^2/c^2 \lesssim 10^{-4}$ derived from offset preservation is weaker than the CMB bound (3.21×10^{-6}) by $\sim 30\times$, so the Bullet Cluster does not improve numerical limits on c_s^2 but does pro-

vide an independent phenomenological confirmation that the Khronon condensate is, in fact, pressureless on scales where it can be tested directly.

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